User Guide: `plot_lagged_coef_panels()`

Source file: `coef_int_plot.R`
Last updated: 2026

Overview

`plot_lagged_coef_panels()` produces a multi-panel base R figure for visualising lagged climate-model coefficients across multiple model fits. It has two regions:

- **Left panels** — one per climate variable, showing each model's point(s) at their lag value on the x-axis and their coefficient on the y-axis.
- **Right panel** — shows interaction and quadratic terms as V-connectors and horizontal lines, linked back to their parent points in the left panels by dashed lines.

The function requires no packages beyond base R.

Quick Start

```r
source("coef_int_plot.R")

coefs <- list(
  base  = coef(fit_base),
  const = coef(fit_const),
  vary  = coef(fit_vary)
)

png("my_figure.png", width = 3700, height = 4050, res = 300)
plot_lagged_coef_panels(
  coefs_named_list = coefs,
  vars_order       = c("nino", "wtio", "etio", "tsa", "aao"),
  main_title       = "Peak Fire-Season (Weeks 51, 52, 1, & 2)",
  model_cols       = c(base = "forestgreen", const = "magenta4", vary = "darkorange2"),
  coef_range       = c(-5, 5),
  add_legends      = TRUE
)
dev.off()
```

Term Naming Rules

This is the most important section. Every coefficient name is parsed by `parse_term()`. Names that do not match a recognised pattern are silently skipped — no error is raised. Use `parse_term()` to check names before plotting.

Kind	Pattern	Example
Main	`<var>_lag<N>`	`nino_lag45`
Interaction	`<var1>_lag<N>:<var2>_lag<M>`	`nino_lag45:etio_lag8`
Quadratic	`I(<var>_lag<N>^2)`	`I(nino_lag45^2)`
Other	anything else	`(Intercept)` — skipped

Variable keys map to display labels as follows:

Key	Label
`nino`	Niño 3.4
`wtio`	WTIO
`etio`	ETIO
`tsa`	TSA
`aao`	SAM
`olr`	OLR
`dmi`	DMI
anything else	uppercased as-is

Common mistakes:

```
parse_term("Nino_lag45")    # "other" - capital N
parse_term("nino_lag_45")  # "other" - extra underscore
parse_term("nino45")        # "other" - no _lag
parse_term("nino_lag45")    # "main"  - correct
```

Interaction anchor requirement

For a V-connector to be drawn, **both sides** of the interaction must appear as main terms in the **same model's** coefficient vector. If either anchor is missing, that connector is silently skipped.

```r
# BAD - etio_lag7 is not a main term in this vector
c(nino_lag45 = 3.5, "nino_lag45:etio_lag7" = -1.2)

# GOOD - both anchors present
c(nino_lag45 = 3.5, etio_lag7 = -4.0, "nino_lag45:etio_lag7" = -1.2)
```

The same rule applies to quadratics: `I(nino_lag45^2)` requires `nino_lag45` to be present as a main term in the same model.

The Coefficient List

`coefs_named_list` is a **named list** of coefficient vectors. The names become the model keys used in every style parameter. You can use any key names you like.

```r
coefs <- list(
  base  = coef(fit_base),   # key = "base"
  const = coef(fit_const),  # key = "const"
  vary  = coef(fit_vary)    # key = "vary"
)
```

The intercept `(Intercept)` is automatically ignored. You do not need to remove it first.

Parameter Reference

Parameters are grouped below by function. All have sensible defaults — a minimal call only needs `coefs_named_list` and `model_cols`.

Data and panels

Parameter	Default	Description
`coefs_named_list`	*(required)*	Named list of coefficient vectors
`vars_order`	`c("nino","wtio","etio","tsa","nao","dtr")`	Which variables get left panels, and in what order. **Omit variables with no terms.**

Parameter	Default	Description
`pch_map`	`c(nino=21, wtio=24, etio=25, tsa=22, aao=23, olr=10)`	Point shape per variable. `olr=10` triggers two-layer rendering (filled disc + crosshair).

Layout

Parameter	Default	Description
`layout_widths`	`c(1.75, 1.25)`	Relative widths of left vs right panel region
`oma`	`NULL`	Outer margins `c(bottom, left, top, right)`. NULL = auto.

Axis ranges

Parameter	Default	Description
`xlim_lag`	`c(52, 1)`	x-axis range for all left panels. The default is reversed so lag 1 plots on the right and lag 52 on the left. Swap to `c(1, 52)` to restore the conventional direction.
`coef_range`	`c(-5, 5)`	y-axis range for all left panels
`coef_range_int`	`coef_range`	x-axis range for the right panel. Can differ from `coef_range`.

Axis ticks

Parameter	Default	Description
`x_axis_at`	`c(52, 40, 30, 20, 10, 1)`	Tick positions on the lag axis. Ordered right-to-left to match the reversed default. Supply any vector to override.
`y_axis_at`	`NULL`	Tick positions on the coefficient axis. NULL = auto from `pretty()` with 0 forced in.

Parameter	Default	Description
`y_axis_las`	`0`	Label orientation for the y-axis only. `0` = parallel to axis, `1` = always horizontal, `2` = perpendicular, `3` = always vertical.
`int_axis_at`	`NULL`	Tick positions for the right panel x-axis. NULL = R default automatic ticks.

Half-ticks

Unlabelled ticks drawn at the midpoint between each pair of adjacent labelled ticks, at half the standard tick length (`tcl = -0.25`). Each axis is controlled independently.

Parameter	Default	Axis
`half_ticks_x`	`FALSE`	Lag axis on all left panels
`half_ticks_y`	`FALSE`	Coefficient axis on all left panels
`half_ticks_int`	`FALSE`	x-axis on the right interaction panel

Titles and text sizes

Parameter	Default	Description
`main_title`	`NULL`	Figure title drawn at `adj=0` inside panel 1's top margin
`cex_main`	`1.75`	`cex` for the title
`title_line`	`1`	`line=` for `title()`. Panel 1's top margin auto-set to `max(2.5, title_line + 1.2)`.
`cex_axis`	`1.2`	`cex.axis` for all axis tick labels
`cex_var_label`	`1.4`	`cex` for the in-panel variable label ("Niño 3.4" etc.)
`var_label_pos`	`5`	Horizontal position of the in-panel variable label, as a **percentage of the lag-axis range** inset from the left visual edge. Works correctly on both normal and reversed axes.

Axis labels

All axis labels are drawn via `mtext()` so their size is fully independent. `cex_lab` is the shared default; the specific parameters override it per label.

Parameter	Default	Description
`xlab_lag`	`"Lag"`	x-axis label on the bottom-most left panel, drawn via `mtext()`
`xlab_coef`	`"Coefficients"`	x-axis label on the right interaction panel. Also used as the fallback for `ylab_left`.
`xlab_coef2`	`NULL`	Optional second line for the right panel x-axis label, drawn below `xlab_coef`. Useful when `xlab_coef` is long at larger `cex_lab_int` values.
`xlab_coef2_line_gap`	`1.2`	Gap in `mtext` line units between `xlab_coef` and `xlab_coef2`. Increase to spread the lines apart; decrease to tighten.
`ylab_left`	`NULL`	Shared outer-margin y-axis label on the left panels. `NULL` falls back to `xlab_coef`. `""` suppresses the label entirely.
`cex_lab`	`1.4`	Default `cex` for all axis labels. Override individually with the parameters below.
`cex_lab_lag`	`NULL`	`cex` for `xlab_lag`. NULL → falls back to `cex_lab`.
`cex_lab_y`	`NULL`	`cex` for `ylab_left`. NULL → falls back to `cex_lab`.
`cex_lab_int`	`NULL`	`cex` for `xlab_coef` and `xlab_coef2` (both lines share the same size). NULL → falls back to `cex_lab`.

Points and lines

Parameter	Default	Description
`cex_pt`	`2.1`	Point size in left panels

Parameter	Default	Description
`cex_pt_int`	`2.0`	Point size in the right panel
`lwd`	`2`	Line width for V-connectors and quadratic lines
`lty_ref`	`2`	Line type for the h=0 / v=0 reference lines
`lwd_ref`	`1.5`	Line width for reference lines

Per-model x-offsets

Parameter	Default	Description
`x_offsets`	`NULL`	Named numeric vector of lag-axis nudges per model key. Separates overlapping points. NULL = all zero. Typical range ±0.25 to ±1.

Per-model colours and styles

All four parameters are **named vectors** keyed by model key. Missing keys fall back to auto-derived defaults silently.

Parameter	Default	Description
`model_cols`	`c(base="forestgreen", const="magenta4", vary="darkorange2")`	Primary colour per model. **Required; names must match `coefs_named_list` keys.**
`model_bgs`	`NULL`	Fill (bg) colour per model. NULL = auto: first model alpha 0.50, rest alpha 0.65 of `model_cols`.
`model_outline`	`NULL`	Outline colour per model. NULL = auto: first model `"grey4"`, rest `"black"`.
`model_lty`	`NULL`	Line type per model (1=solid, 2=dashed, 3=dotted…). NULL = all dashed (2).

Jitter

`int_y_jitter` and `quad_y_jitter` are handled by a single unified system. When an interaction arm and a quadratic line share the same anchor point, they are offset together as one group so they can never overlap regardless of which jitter values are set. The direction logic works as follows:

- **Interaction arms** are assigned the panel index of their *other-side* variable in `vars_order`. An arm connecting toward a panel above the anchor (smaller index) receives a positive offset; one connecting below (larger index) receives a negative offset.
- **Quadratic lines** are assigned the panel index of the *anchor itself*. This places them between arms heading above and arms heading below: if the interaction goes downward the quadratic is nudged up; if the interaction goes upward the quadratic is nudged down.
- A single departure from an anchor always gets offset 0.

Parameter	Default	Description
`quad_y_jitter`	`0.10`	y-separation step for quadratic lines when sharing an anchor with an interaction arm. Scaled by the same directional multiplier as `int_y_jitter`.
`int_y_jitter`	`0`	y-separation step for interaction arms sharing an anchor. Arms toward a higher panel receive a positive offset; arms toward a lower panel receive a negative offset. Both this and `quad_y_jitter` must be non-zero for mixed-anchor separation to be visible.
`int_x_jitter`	`0`	x-shift of the vertical bar per unique interaction term. Horizontal arms terminate at the shifted x, keeping the T-junction geometry intact.

Auto-jitter for overlapping left-panel points

When two or more models have effectively the same lag and coefficient value, one point will sit directly on top of another. Enable `auto_jitter` to spread them apart automatically.

Parameter	Default	Description
`auto_jitter`	`FALSE`	Master switch. When `TRUE`, overlapping points are detected and nudged before drawing.
`auto_jitter_x`	`0.4`	x nudge step size in lag units. Each point in an overlap group is offset by a symmetric multiple of this value.
`auto_jitter_y`	`0.0`	y nudge step size in coefficient units. Set to a small value (e.g. `0.05`) if points are also identical in coefficient value.
`auto_jitter_tol_x`	`1.5`	Tolerance in lag units for declaring two points overlapping. Points within this distance in x **and** `auto_jitter_tol_y` in y are grouped.
`auto_jitter_tol_y`	`0.3`	Tolerance in coefficient units for overlap detection.

The nudges are applied to both the drawn points and the NDC anchors stored for the right panel, so V-connector and quadratic linking lines follow the nudged positions exactly. Typical usage: `auto_jitter = TRUE, auto_jitter_x = 0.4`. Add `auto_jitter_y = 0.05` if models also share very similar coefficient values.

Auto x-jitter for overlapping interaction coefficients

When multiple interaction terms have nearly identical coefficient values in the right panel, their V-connectors stack on top of each other. Enable `auto_int_x_jitter` to spread them apart along the x-axis.

Parameter	Default	Description
`auto_int_x_jitter`	`FALSE`	Master switch. When `TRUE`, `(model, iterm)` pairs whose interaction coefficient values fall within `auto_int_x_tol` are automatically nudged apart.
`auto_int_x_nudge`	`0.10`	x nudge step size in coefficient units.

Parameter	Default	Description
`auto_int_x_tol`	`0.15`	Tolerance in coefficient units for declaring two interaction values as overlapping.

The nudge is applied to `xint_j` — the single x value from which all four segments and the star marker of each V-connector are drawn — so the entire connector shifts together automatically.

Y-axis label for left panels

Parameter	Default	Description
`ylab_left`	`NULL`	Shared outer-margin y-axis label on the left panels. `NULL` falls back to `xlab_coef`. `""` suppresses the label entirely. Any other string is used verbatim.

Legends

Parameter	Default	Description
`add_legends`	`FALSE`	Draw the Terms and Model legends in the right panel
`legend_inset_terms`	`c(0.000, 0.00)`	`inset=` for the Terms legend `c(h_from_right, v_from_top)`
`legend_inset_model`	`c(0.000, 0.21)`	`inset=` for the Model legend. Increase `[2]` to push it below the Terms box.
`legend_cex_terms`	`2.25`	`cex` for Terms legend (sized for 300 dpi PNG)
`legend_cex_model`	`2.0`	`cex` for Model legend
`legend_pt_bg`	`alpha_col("gray32", 0.65)`	`pt.bg` for Terms legend symbols
`legend_terms`	`c("Niño 3.4","WTIO","ETIO","TSA","SAM","OLR","Interaction")`	Term labels. Trim to match what's actually in the figure.
`legend_terms_pch`	`c(21,24,25,22,23,10,11)`	pch vector matching `legend_terms`

Parameter	Default	Description
`legend_terms_pt_cex`	`c(2.25,1.8,1.8,2.25,2.25,2.25,1.8)`	pt.cex vector matching `legend_terms`
`legend_models`	`c("All-Data","Fixed-Selection","Withheld-Season")`	Model labels
`legend_model_keys`	`c("base","const","vary")`	Keys into `model_cols` for the colour swatches
`legend_model_pch`	`15L`	Point symbol in the Model legend. The model's line type (from `model_lty`) is always drawn alongside it, giving a type-"b" appearance.

Helper Functions

These are exported alongside the main function and can be called directly.

`parse_term(term)`

Classifies a single coefficient name. Returns a list with `$kind` and other fields depending on the kind.

```
parse_term("nino_lag45")            # $kind = "main", $var = "nino", $lag = 45
parse_term("nino_lag45:etio_lag8") # $kind = "interaction"
parse_term("I(nino_lag45^2)")       # $kind = "quad", $base_term = "nino_lag45"
parse_term("(Intercept)")           # $kind = "other"
```

`coef_to_df(coef_vec, model_name)`

Converts a named coefficient vector into a tidy data frame. Useful for inspecting how terms will be classified before plotting.

```
df <- coef_to_df(c(nino_lag45 = 4.2, "nino_lag45:etio_lag8" = -1.9), "base")
print(df[, c("term", "kind", "var", "lag")])
```

`build_model_styles(model_cols, ...)`

Pre-resolves all per-model visual properties into a named list. Useful for inspecting auto-derived values or making fine-grained overrides before passing back in.

```r
styles <- build_model_styles(
  model_cols = c(base = "forestgreen", const = "magenta4"),
  x_offsets  = c(base = -0.25, const = 0.25)
)
str(styles)
# List of 2
#  $ base :List of 5  $ col $ bg $ outline $ lty $ xoff
#  $ const:List of 5  ...

# Override one bg, then extract for passing back in:
styles$const$bg <- adjustcolor("magenta4", alpha.f = 0.9)
manual_bgs <- sapply(styles, `[[`, "bg")
# plot_lagged_coef_panels(..., model_bgs = manual_bgs)
```

`alpha_col(col, a)`

Thin wrapper around `adjustcolor()`. Used internally but available for consistency when building manual colour vectors.

Common Pitfalls

1. Wrong term names. Always run `parse_term()` on a few names first. Capital letters, extra underscores, or missing `_lag` will silently produce `kind = "other"` and the term will not appear.

2. Interaction anchor missing. If a V-connector doesn't appear, check that both the `left_term` and `right_term` are present as main terms in the same model's coefficient vector.

3. `vars_order` includes a variable with no terms. This produces a blank left panel. Remove that key from `vars_order`.

4. Model key mismatch. Keys in `model_cols`, `x_offsets`, `model_lty`, etc. must match the names in `coefs_named_list`. Mismatches fall back to defaults without an error.

5. Legend boxes overlapping. Increase `legend_inset_model[2]` by ~0.03–0.05 per step until the Model box clears the bottom of the Terms box. The exact value depends on `legend_cex_terms` and the number of entries.

6. Crowded right panel — lines overlapping. For separation of lines in the right panel, `int_y_jitter` and `quad_y_jitter` work together — set both to non-zero values when an interaction and a quadratic share the same anchor (e.g. `int_y_jitter = 0.03, quad_y_jitter = 0.10`). For nearly coincident interaction coefficient values, use `auto_int_x_jitter = TRUE`. Use `int_x_jitter = 0.05–0.15` to manually separate vertical bars of different interaction terms.

7. Auto-jitter tolerance too wide or too narrow. If `auto_jitter` is grouping left-panel points that shouldn't move together, tighten `auto_jitter_tol_x` and `auto_jitter_tol_y`. The defaults (`tol_x = 1.5` lag weeks, `tol_y = 0.3` coefficient units) work well when models share the same nominal lag.

8. Two-line label clipping. If `xlab_coef2` is still clipping at large `cex_lab_int` values, increase the bottom margin of the right panel by adding margin space via `oma`.

Legend Spacing Formula

The two legends ("Terms" and "Model") are stacked by adjusting `legend_inset_model[2]`. A rough starting formula:

```
legend_inset_model[2]   0.04 × length(legend_terms) × (legend_cex_terms / 2.25)
```

Then fine-tune by eye. At `legend_cex_terms = 2.25` with 7 terms, a value around `0.21` works well.